

Structured Reseeding After Failure: Asymmetric Selection–Transformation Cycles in Macroeconomic, Biological, and Cognitive Systems

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Abstract

When exploitation fails, standard decision-making models predict that systems explore again—either randomly, proportional to uncertainty, or toward novel options. We report evidence that post-failure reorientation is fundamentally more structured than these models predict. In a four-armed restless bandit task (20 participants, 16,000 trials), human decision-makers who abandon a failing option target alternatives unvisited for a median of 15 trials—versus 2 trials during routine exploration—and return to the abandoned option within 4 trials (77.8% of events; $p = 3.9 \times 10^{-83}$, universal across all 20 participants). This targeted-reconnaissance-and-return pattern is not predicted by upper confidence bound, Thompson sampling, or optimal foraging models.

This reseeded signature generalizes across scales. In macroeconomic data (FRED, 1947–2025), R&D investment growth precedes capacity utilization changes with strong asymmetry ($AI = -0.92$) but not the reverse. In GPS-tracked avian foraging (84 lesser black-backed gulls), search and exploitation phases are structurally anti-correlated ($\rho = -0.75$, 100% of individuals negative, Cohen's $d = -8.57$), with unidirectional search-to-exploitation flow ($p = 0.018$). Mammal movement data replicate this asymmetry ($p < 0.002$).

We interpret these findings through the The First Pattern (TFP) framework, which predicts unidirectional selection-to-transformation flow with viability-gated reseeded: when exploitation fails, the system enters a brief reorientation mode (Q4B) that targets novel contexts before cycling back. The distinction between initial exploration and post-failure reseeded—supported across economic, biological, and cognitive domains—suggests that adaptive information processing follows a directed lifecycle architecture rather than a symmetric explore–exploit tradeoff.

Keywords: adaptive decision-making, explore–exploit tradeoff, temporal foraging policy, viability gating, reseeded, cross-domain validation, information lifecycle

1. Introduction

1.1 The Post-Failure Problem in Adaptive Decision-Making

A fundamental challenge for any adaptive system—biological, cognitive, or economic—is how to respond when a currently exploited strategy fails. Standard theoretical frameworks offer several answers. Optimal foraging theory (OFT; Stephens & Krebs, 1986) predicts that organisms leave depleted patches and resume search, treating post-failure search as functionally equivalent to initial exploration. Reinforcement learning models employing upper confidence bound (UCB; Auer et al., 2002) or Thompson sampling (Thompson, 1933) allocate exploration proportional to uncertainty, directing attention toward options with uncertain value estimates. Epsilon-greedy strategies (Sutton & Barto, 2018) switch randomly between exploitation and exploration at a fixed rate, with no structural distinction between initial and post-failure search.

Even hierarchical reinforcement learning, which distinguishes between intra-option policies and a meta-policy that selects among options (Sutton et al., 1999), treats post-termination option selection as governed by the same meta-policy used for initial selection—it does not predict structured reseeding with targeted novelty and systematic return.

These models share a critical assumption: that the transition from exploitation back to exploration is symmetric, or at least functionally equivalent to the initial exploration phase. When exploitation fails, the system simply "explores again."

1.2 An Alternative: Structured Reseeding

The First Pattern (TFP) framework (Ollar, 2026a, 2026b, 2026c) offers a structurally distinct prediction. TFP proposes that adaptive information processing follows a directed lifecycle with four functional phases: internal orientation (Q1), external selection (Q2), internal transformation and external commitment (Q3), and outward deployment (Q4). Critically, deployment bifurcates into continuation while viable (Q4A) and viability-gated reorientation when the current strategy collapses (Q4B).

The TFP prediction that distinguishes it from standard models is this: Q4B (post-failure reorientation) is not equivalent to Q2 (initial selection). Rather, Q4B constitutes a distinct functional mode—a reseeding operation—that routes through a brief reorientation before initiating a new lifecycle. Furthermore, the cycle is strictly unidirectional: $Q2 \rightarrow Q3$ (selection leads to commitment), but Q3 never directly returns to Q2. Failures at any stage route through Q4B before re-entering Q1, creating an asymmetric temporal structure.

1.3 Specific Predictions

We derive five falsifiable predictions from TFP that are not jointly predicted by OFT, UCB, Thompson sampling, or threshold-based switching models:

P1 (Structural Opposition): Selection (Q2) and transformation (Q3) are structurally anti-correlated—when a system is in selection mode, it is not in transformation mode, and vice versa.

P2 (Unidirectional Flow): Q2 precedes Q3 asymmetrically. Cross-correlation between selection and transformation signals should be negative at positive lags (Q2 leads Q3) but near zero at negative lags (Q3 does not directly lead Q2).

P3 (Viability Gating): The transition from exploitation back toward exploration is triggered by viability collapse (sustained performance decline), not by stochastic switching or uncertainty accumulation.

P4 (Structural Openness and Contextual Anchoring): Post-failure reorientation (Q4B) creates a brief structural openness—a pause in which the system becomes available to its local environment. This openness enables immediate contextual anchoring (Q1), in which the system locks onto a novel alternative based on local availability rather than directed search. The resulting selection should favor options that are systematically more novel than those chosen during routine exploration (Q2).

P5 (Reseeding Loop Completion): After contextual anchoring, the system returns to previously abandoned options, completing a reseeding loop (Q4B → Q1 → Q2 → Q3). This return pattern distinguishes TFP from permanent abandonment models.

1.4 Scope

We test these predictions across four domains: macroeconomics (quarterly data spanning 78 years), avian foraging behavior (GPS tracking of 84 individuals), mammal movement ecology (GPS tracking), and human sequential decision-making (multi-armed bandit). We validate the transformation–reseeding segment of the TFP lifecycle (Q2 → Q3 → Q4B → Q1 → Q2), with indirect evidence for Q1 via the novel-option anchoring that immediately follows Q4B openness. Full direct validation of Q1 (internal orientation) and Q4A (propagation continuation) requires richer process measures (e.g., eye-tracking, neural recording) and remains future work.

2. Theoretical Framework

2.1 The First Pattern

TFP describes adaptive information processing as a directed lifecycle comprising four quadrants, each defined by a pair of information operators:

- Q1 (Prime): Bind_in → Explore_in. Internal stabilization followed by internal variation. The system centers itself and generates potential directions. Analogous to gravitational centering in physics or preparation phases in cellular replication.
- Q2 (Select/Bind): Explore_out → Bind_in. External scanning followed by internal consolidation. The environment selects viable candidates and the system locks onto them. Analogous to the strong nuclear force binding selected particles, or the egg selecting compatible sperm.
- Q3 (Fuse/Transform): Explore_in → Bind_out. Internal recombination followed by external commitment. Separate elements fuse into a new structural whole, which is then actualized externally. Analogous to weak nuclear force enabling transformation, or genetic fusion in fertilization.
- Q4 (Exchange/Deploy): Bind_out → Explore_out. External commitment followed by outward propagation.
- Q4A (Copy): Successful deployment continues. The pattern replicates.
- Q4B (Reorient): Viability collapses. Bind_out carries the structural residue of the failed deployment,

and Explore_out creates a brief moment of openness—a structural availability to the local environment. This is not directed search but rather the creation of conditions under which a new Q1 can naturally form. Q4B brings the "tools and mechanisms" that enable environmental conditions where Q1 (contextual anchoring) emerges almost immediately.

2.2 Process vs. Contextual Quadrants

A key structural distinction within TFP is that Q2 and Q3 are process quadrants—their activation is behaviorally salient and directly observable as probing and committing actions. Q1 and Q4 are contextual quadrants—they establish conditions for processes but produce weaker, more diffuse behavioral signatures. Each quadrant comprises two information operators (an entry operator and an exit operator) that together produce the quadrant's function; the operators do not function independently. This asymmetry predicts that $Q2 \leftrightarrow Q3$ opposition should be empirically strong and detectable, while $Q1 \leftrightarrow Q4$ dynamics may be below measurement thresholds in behavioral data.

2.3 Distinction from Standard Explore-Exploit Models

Standard explore–exploit theories predict symmetric or uncertainty-proportional transitions between exploration and exploitation. TFP makes three structurally distinct predictions:

1. Asymmetric temporal flow: $Q2 \rightarrow Q3$ is fast and direct; $Q3 \rightarrow Q2$ can only occur via $Q4B \rightarrow Q1$, introducing a mandatory intermediary.
2. Viability gating: The $Q3 \rightarrow Q4B$ transition requires a viability collapse signal, not merely the passage of time or accumulation of uncertainty.
3. Targeted novelty with return: Q4B does not explore randomly or proportional to uncertainty. It targets specifically novel options (reseeding into new contexts) and then returns to the abandoned option, completing a reconnaissance-and-return loop.

No existing model in optimal foraging theory, multi-armed bandit literature, or drift-diffusion frameworks jointly predicts all three of these features.

3. Methods

3.1 Data Sources

Domain 1: Macroeconomics. Quarterly time series from the Federal Reserve Economic Data (FRED) database, spanning 1947–2025 ($N = 315$ quarters). Q2 proxy: gross domestic R&D investment growth rate (series Y006RC1Q027SBEA). Q3 proxy: total capacity utilization change (series TCU). Both series were first-differenced to achieve stationarity and z-scored.

Domain 2: Avian Foraging. GPS tracking data from 84 lesser black-backed gulls (*Larus fuscus*) via the Movebank Data Repository (Wikelski et al., 2015; doi:10.5441/001/1.q986rc29). Total: 89,867 GPS fixes. Q2 proxy: rolling fraction of time with velocity above the individual median (search intensity). Q3 proxy: rolling fraction of time with velocity below the 25th percentile and low acceleration (exploitation intensity). Window size: 10 consecutive fixes.

Domain 3: Mammal Movement. GPS tracking of one domestic horse from the Movebank Data Repository (Fischer, University of Copenhagen). Total: 17,517 fixes. Same Q2/Q3 operationalization as avian data.

Domain 4: Human Decision-Making. Published behavioral data from Speekenbrink and Konstantinidis (2015): 20 participants performing a four-armed restless bandit task across four experimental conditions (N = 16,000 trials). Q2 proxy: arm-switching trials (exploration). Q3 proxy: arm-staying trials following a switch (exploitation). Q4B events: exploitation-to-switch transitions preceded by reward decline below 50% of the recent five-trial moving average. Data accessed from <https://github.com/speekenbrink-lab/data>.

3.2 Analytical Approach

Structural Opposition (P1). For GPS data (Domains 2–3), we computed Pearson correlation between Q2 and Q3 intensity signals at lag 0 for each individual. For human data, we computed transition probabilities $P(Q3|Q2)$ and $P(Q2|Q3)$.

Unidirectional Flow (P2). We computed cross-correlation functions $CC(Q2(t), Q3(t+\tau))$ for $\tau \in [-20, +20]$ in all four domains. We defined the Asymmetry Index as $AI = [\rho(\tau=+k) - \rho(\tau=-k)] / [\max(|\rho|) + 0.001]$ for $k \in \{2, 3\}$. Significance was assessed via permutation testing (N = 500 shuffles for animal data, N = 1000 for economic data) and paired t-tests for human within-participant analyses.

Viability Gating (P3). In human data, we compared the mean reward on the trial immediately preceding Q3 → Q3 transitions (exploitation continues) versus Q3 → Q4B transitions (exploitation abandoned). We assessed significance with a two-sample t-test.

Targeted Redirection (P4). For each Q4B event in human data, we computed the number of trials since the agent last visited the target arm. We compared this recency distribution against fresh Q2 transitions using the Mann–Whitney U test and computed the rank-biserial correlation as an effect size measure. We validated consistency with a within-participant Wilcoxon signed-rank test.

Reseeding Loop Completion (P5). For each Q4B event, we tracked whether the participant returned to the abandoned arm within 50 subsequent trials and recorded the lag to return.

3.3 Alternative Model Predictions

We explicitly compared our findings against predictions from: - Upper Confidence Bound (UCB): Predicts exploration proportional to uncertainty. Post-failure transitions should target high-uncertainty options (recently abandoned arms have moderate uncertainty; novel arms have high uncertainty). Does not predict systematic return to abandoned options. - Thompson Sampling: Predicts exploration proportional to posterior probability of optimality. Does not predict novelty targeting or return behavior. - Simple Threshold Models: Predict switching when reward falls below a fixed or moving threshold. Do not predict arm-selection targeting or return dynamics. - Optimal Foraging Theory: Predicts patch-leaving when marginal value falls below environmental average. Does not specify post-leaving search structure or return behavior.

4. Results

4.1 Targeted Reseeding After Viability Collapse (Finding 4: P4)

The most striking finding concerns the structure of post-failure reorientation in human decision-making. We identified 230 Q4B events (exploitation-to-switch transitions following reward decline) and 2,504 fresh Q2 events (exploration transitions not preceded by exploitation failure) across 80 sessions.

Q4B transitions targeted arms that had not been visited for a median of 15 trials, compared to a median of 2 trials for fresh Q2 transitions ($U = 479,214$, $p = 3.9 \times 10^{-83}$, rank-biserial $r = -0.726$). This effect was universal: all 20 participants showed higher target novelty for Q4B than Q2 (Wilcoxon $W = 0$, $p = 1.9 \times 10^{-6}$).

Notably, post-Q4B exploration was not broader than fresh Q2 in terms of arm diversity (mean unique arms in 5 subsequent trials: Q4B = 2.07, Q2 = 2.16, $p = 0.062$). Q4B reseeded is targeted, not diffuse—consistent with TFP's prediction of directed reorientation into a specific novel context rather than broad uncertainty-reduction.

Q4B transitions accepted a significant short-term reward cost (mean post-event reward: Q4B = 8.98, Q2 = 22.99, $p = 1.2 \times 10^{-16}$), indicating that the reseeded function prioritizes contextual novelty over immediate payoff.

These patterns are consistent with a two-step TFP process: a Q4B structural openness phase, reflected in the elevated response time (Section 4.5), followed by immediate Q1 contextual anchoring to locally available novel options. This provides indirect evidence for Q1 as a distinct phase: the system does not search broadly (arm diversity is equivalent to fresh Q2) but rather anchors rapidly to a specific novel context—consistent with Bind_in locking onto what is environmentally available rather than Q4B conducting directed search.

These patterns violate UCB predictions (which would target high-uncertainty arms, including recently abandoned ones) and Thompson sampling predictions (which would sample proportional to expected value, not recency-weighted novelty). Unlike these models, which predict continuous value-driven selection, TFP predicts a discrete temporal sequence: viability collapse (Q4B) → contextual reset (Q1) → reselection (Q2). The 90ms RT elevation is not predicted by UCB, in which switching is immediate value comparison without structural pause.

4.2 Reseeding Loop Completion (Finding 5: P5)

Consistent with TFP's prediction of a reseeded loop (Q4B → Q1 → Q2 → Q3), 77.8% of Q4B events were followed by a return to the specific abandoned arm within 50 trials, with a median return lag of 4 trials. Participants returned to exactly the same option they had abandoned—accepting immediate suboptimality—rather than merely sampling any previously visited option. This targeted reconnaissance-and-return loop is not predicted by any standard decision-making model: UCB would not predict return to a low-reward arm (uncertainty alone is insufficient when expected value has collapsed), and threshold models specify departure but not destination or return.

The four-trial return latency suggests that the reseeding operation is brief and focused: the system makes a short excursion into novel territory, gathers information or resets internal state, and then re-evaluates the previously abandoned option under updated conditions. This is consistent with TFP's characterization of Q4B as a "faint" transitional mode—energetically minimal but functionally essential.

4.3 Unidirectional Selection-to-Transformation Flow (Finding 2: P2)

Across all four domains, selection (Q2) preceded transformation (Q3) asymmetrically.

Economics: R&D investment growth preceded capacity utilization changes with an asymmetry index of $AI(lag=3) = -0.92$ ($N = 315$ quarters). R&D increases predicted subsequent capacity expansion, but capacity changes did not predict R&D shifts at equivalent lags.

Avian Foraging: In 84 lesser black-backed gulls, mean asymmetry index was $+0.022$ (t-test: $p = 0.018$; permutation test: $p < 0.002$; 0/500 shuffled datasets exceeded observed AI). Sixty percent of individual birds showed positive asymmetry.

Mammal Movement: In horse GPS data, $AI = +0.017$ (permutation $p < 0.002$).

Human Decision-Making: The transition probability ratio $P(Q3|Q2) / P(Q2|Q3) = 3.2$ ($t(79) = 13.44$, $p < 10^{-20}$). Participants were 3.2 times more likely to transition from exploration to exploitation than the reverse.

The consistency of this asymmetry across quarterly macroeconomic cycles, hourly GPS trajectories, and second-level human decisions suggests a scale-invariant structural property of adaptive information flow.

4.4 Structural Opposition Between Selection and Transformation (Finding 1: P1)

In avian GPS data, Q2 (search intensity) and Q3 (exploitation intensity) were strongly anti-correlated at lag 0: mean $\rho = -0.75$ (Cohen's $d = -8.57$). All 84 individual birds showed negative Q2–Q3 correlation (100% consistency). The operational definitions ensured non-trivial complementarity: on average, 25.2% of each bird's time was spent in a "neither" zone (neither high-velocity search nor low-velocity stationary exploitation), confirming that the anti-correlation reflects genuine process opposition rather than a measurement tautology.

4.5 Viability Gating (Finding 3: P3)

In human bandit data, the mean reward on the trial preceding exploitation continuation ($Q3 \rightarrow Q3$) was 43.87, compared to 15.63 on the trial preceding exploitation abandonment ($Q3 \rightarrow Q4B$)—a decline of 28.24 points ($p < 0.001$). This is consistent with TFP's prediction that Q4B entry is gated by a viability signal rather than by random switching, uncertainty accumulation, or fixed temporal schedules.

Response time data provide additional evidence for a distinct Q4B mode: Q4B-triggered switches were slower than fresh Q2 switches (2,111 ms vs. 2,023 ms, $p < 0.001$), consistent with a deliberative reorientation process rather than habitual exploration.

4.6 Synthetic Validation

To test whether TFP structure emerges in successful agents independently of task design, we trained unconstrained PPO-LSTM agents (no TFP architecture) on a diagnostic POMDP (ClinicGrid). In successfully trained agents, Q2–Q3 anti-correlation was -0.507 , while random-policy agents showed no anti-correlation ($TSAS = -1,147$, indicating generic models fit random data better). This circularity control confirms that TFP-like structure is a property of successful adaptive behavior, not a fitting artifact.

5. Discussion

5.1 Summary

We present cross-domain evidence for three structural properties of adaptive decision-making predicted by TFP and not jointly predicted by standard explore–exploit models: (1) anti-correlation between selection and transformation processes, (2) unidirectional forward flow from selection to transformation, and (3) viability-gated reseeding with targeted redirection and systematic return. The most novel finding—that post-failure reorientation targets specifically novel options and includes a rapid return to abandoned alternatives ($p = 3.9 \times 10^{-83}$, universal across participants)—constitutes a behavioral signature not predicted by UCB, Thompson sampling, optimal foraging, or threshold-based switching.

5.2 Alternative Theories

We consider four classes of alternative models and evaluate whether each can account for the full behavioral profile we observe: viability-gated switching (P3), targeted novelty selection (P4), and systematic return to abandoned options (P5).

Optimal foraging theory (OFT). ** The marginal value theorem (Charnov, 1976) predicts patch-leaving when the marginal intake rate within a patch falls below the average rate across all patches. Our viability-gating finding (P3: reward decline from 43.87 to 15.63 triggers switching) is compatible with this prediction. However, OFT specifies *when* to leave but not *where* to go: it does not predict that post-leaving search should target specifically novel options (P4: arms unvisited for 15 trials) rather than any available alternative. More critically, OFT does not predict return to the abandoned patch within a characteristic timeframe (P5: 77.8% return within 4 trials). The marginal value theorem addresses patch residence time, not the temporal structure of post-departure behavior. **OFT accounts for P3 but not P4 or P5.

Upper confidence bound (UCB) algorithms. ** UCB agents allocate exploration proportional to the uncertainty of each option (Auer et al., 2002). After abandoning a failing arm, that arm retains moderate uncertainty (it was recently sampled), while arms not sampled for many trials accumulate high uncertainty. UCB therefore *does* predict exploration of novel arms (partially consistent with P4). However, UCB predicts continued sampling of high-uncertainty arms until uncertainty is reduced—it does not predict a *rapid return* to the recently abandoned arm (P5). In our data, participants target novel arms (median 15 trials unvisited) but return to the abandoned arm within 4 trials, a pattern

inconsistent with continued uncertainty reduction. Furthermore, UCB does not distinguish between post-failure exploration and routine exploration: the same uncertainty-based rule governs both. Our data show that Q4B transitions are behaviorally distinct from Q2 transitions (slower RT, $p < 0.001$; different target recency, $p = 3.9 \times 10^{-83}$). **UCB partially accounts for P4 but not P5, and does not predict Q4B/Q2 behavioral distinction.

Drift-diffusion models (DDMs). ** DDMs accumulate evidence toward a decision boundary (Ratcliff & McKoon, 2008). After boundary crossing (commitment to an option), the model resets. DDMs can potentially account for viability gating through a declining evidence signal. However, they do not specify the *content* of post-reset exploration—what options are explored, in what order, or whether return to abandoned options occurs. The targeted novelty and return behavior we observe is outside the scope of single-decision DDMs. **DDMs do not address P4 or P5.

Simple reward-threshold models. ** Threshold switching (abandon when reward $< \theta$) accounts for our viability-gating finding (P3) but does not predict which arm is selected next (P4) or whether return occurs (P5). Threshold models specify a *departure rule* but not a *destination rule*. **Threshold models account for P3 only.

Hierarchical reinforcement learning (HRL). ** HRL with options (Sutton et al., 1999) distinguishes between intra-option policies and a meta-policy that selects among options. When an option terminates (due to low reward), the meta-policy selects a new option. This is structurally the closest standard model to TFP. However, HRL does not predict (a) that post-termination option selection should systematically favor novel options over recently visited ones, or (b) that a return to the terminated option should occur within a characteristic timeframe. HRL *could* implement TFP-like structure, but this would require hand-coding the termination condition (viability collapse triggers meta-controller activation) and the reseeding stack (return-to-origin pointer). TFP predicts these structures as intrinsic properties of adaptive lifecycles; standard HRL treats them as engineering choices. Our synthetic validation (ClinicGrid) shows that TFP-like asymmetry *emerges* in unconstrained agents without architectural imposition, and our cross-domain data show it persists from quarterly economic cycles to second-level human decisions—suggesting the structure is intrinsic, not engineered. **HRL is the strongest alternative but does not predict the return dynamics without hand-coding.

In summary, each alternative model accounts for a subset of our findings. Only TFP jointly predicts viability-gated switching (P3), targeted novelty selection (P4), systematic return to abandoned options (P5), and the behavioral distinction between initial exploration and post-failure reorientation. The reconnaissance-and-return signature—specifically, the 4-trial median return latency with cost acceptance—is the empirical feature that most clearly differentiates TFP from existing frameworks.

5.3 Scope and Limitations

Validated: We provide evidence for the process-level core of TFP—specifically, the $Q2 \rightarrow Q3 \rightarrow Q4B \rightarrow Q1 \rightarrow Q2$ reseeding loop. The structural opposition of Q2 and Q3, the unidirectional

forward flow, and the viability-gated reseeding mechanism with targeted redirection and return are supported across domains.

Indirectly validated: Q1 (internal orientation/contextual anchoring) receives indirect support from the novel-arm anchoring data: the rapid, non-diffuse selection of a specific novel alternative immediately following Q4B openness is consistent with Bind_in locking onto locally available environmental conditions. However, direct measurement of Q1's internal operators (Bind_in stabilization, Explore_in direction formation) awaits process-tracing studies with richer measures (confidence ratings, pupillometry, or neural recording). While our data cannot temporally resolve whether novel-arm anchoring occurs during Q4B openness or immediately post (Q1), the behavioral profile—targeted rather than diffuse, brief rather than extended—is more consistent with contextual anchoring (Q1) than directed search (Q2).

Not validated: Q4A (propagation continuation) is not directly observed in the present study. Its validation requires multi-stage deployment tracking beyond the current data. We therefore validate the transformation–reseeding segment of the TFP lifecycle with indirect Q1 evidence, not the complete cycle. This is an explicit scope limitation, not a hidden weakness.

Effect size variation: Asymmetry indices ranged from -0.92 (economics, quarterly) to $+0.017$ (horse GPS, minutes-level). We interpret this variation as reflecting signal-to-noise differences across temporal scales rather than theory inconsistency: coarsely sampled data reveals structural asymmetry more cleanly, while fine-grained GPS data preserves the directional signal despite micro-movement noise.

Single human dataset: Our human findings derive from one published dataset (Speekenbrink & Konstantinidis, 2015). Replication in additional human decision-making paradigms remains essential.

Causality: Our evidence is structural and correlational. We demonstrate that adaptive systems exhibit TFP-consistent temporal organization, but we do not manipulate the lifecycle structure. Causal tests—such as engineering agents with and without Q4B reseeding capability and comparing resilience to viability shocks—are a priority for future work.

5.4 Implications

For cognitive science: The Q4B reseeding loop suggests that human post-failure exploration is not simply "exploring again" but involves a structured reconnaissance-and-return operation. Decision-making models should distinguish between initial exploration (Q2) and post-failure reorientation (Q4B), as these produce measurably different behavioral signatures (novelty targeting, cost acceptance, return dynamics).

For movement ecology: GPS analyses of animal foraging should distinguish between initial search bouts and post-depletion reorientation bouts, as these may serve different functional roles in the foraging lifecycle. Our finding that search and exploitation anti-correlate at the individual level (100% of 84 gulls) suggests that dual-mode classification is behaviorally meaningful in movement ecology.

For economics: Business cycle models may benefit from incorporating a reseeding phase distinct from both expansion and simple contraction. The strong unidirectional asymmetry from R&D

investment to capacity utilization ($AI = -0.92$) suggests that innovation-to-deployment flow has a characteristic directionality that could inform investment timing models.

For artificial intelligence: RL agents designed with a TFP-inspired Q4B module—viability monitoring, targeted novelty selection, and return-to-abandoned-option stacking—may show improved resilience to non-stationary environments compared to standard UCB or epsilon-greedy exploration.

5.5 Falsifiability

The findings reported here could have falsified TFP at multiple points. Had the cross-correlation between Q2 and Q3 been symmetric (equal strength at positive and negative lags), TFP's unidirectional flow prediction would have failed, and the data would have been consistent with standard symmetric explore–exploit models. Had Q4B-triggered arm selections shown the same recency distribution as fresh Q2 selections, TFP's claim of a distinct reseeding mode would have been indistinguishable from generic post-failure exploration. Had participants not returned to abandoned arms—or returned at random latencies rather than a characteristic 4-trial window—the reseeding loop prediction would have been unsupported. None of these falsification conditions obtained. In each case, the data conformed to TFP's specific predictions while violating the predictions of at least one standard alternative.

5.6 Future Directions

Three extensions are immediate priorities. First, direct measurement of Q1 (internal orientation) and Q4A (deployment continuation) using process-tracing methods in human decision-making. Second, causal intervention studies comparing TFP-structured agents against standard RL baselines in viability-shock environments. Third, expansion to additional domains—particularly immune cell behavior (where the search → conjugate → kill → release cycle provides a biological analog) and developmental biology (where lifecycle phases are independently documented).

6. Conclusion

Adaptive systems do not merely explore after failure; they reseed. Across macroeconomic cycles, avian foraging, mammal movement, and human bandit decisions, the evidence presented here suggests that these systems employ structured reorientation rather than stochastic search when exploitation fails. Instead, post-failure reorientation involves a structured reseeding operation: targeted novelty selection, short-term cost acceptance, and systematic return to abandoned options. This viability-gated reseeding loop—predicted by the The First Pattern framework and not jointly predicted by existing explore–exploit models—suggests that the distinction between initial exploration and post-failure reorientation is a fundamental structural property of adaptive information processing across scales.

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Data Availability

All datasets used in this study are publicly available. FRED economic data: <https://fred.stlouisfed.org>. Movebank GPS data: <https://www.movebank.org> (doi:10.5441/001/1.q986rc29). Human behavioral data: <https://github.com/speekenbrink-lab/data>. Analysis scripts and results are available at [repository URL upon publication].

Author Contributions

A.O. conceived the theoretical framework, designed the analyses, conducted all computational work, and wrote the manuscript.

Use of AI Tools

The author used AI assistants (Claude, GPT 5.4) as tools for code development, figure generation, and help with drafting and editing prose. All study design, experiments, data analysis, and scientific claims were specified, checked, and approved by the author, who takes full responsibility for the content of this manuscript.

Competing Interests

The author declares no competing interests.

The First Pattern™ Map: Information Emergence Cycle

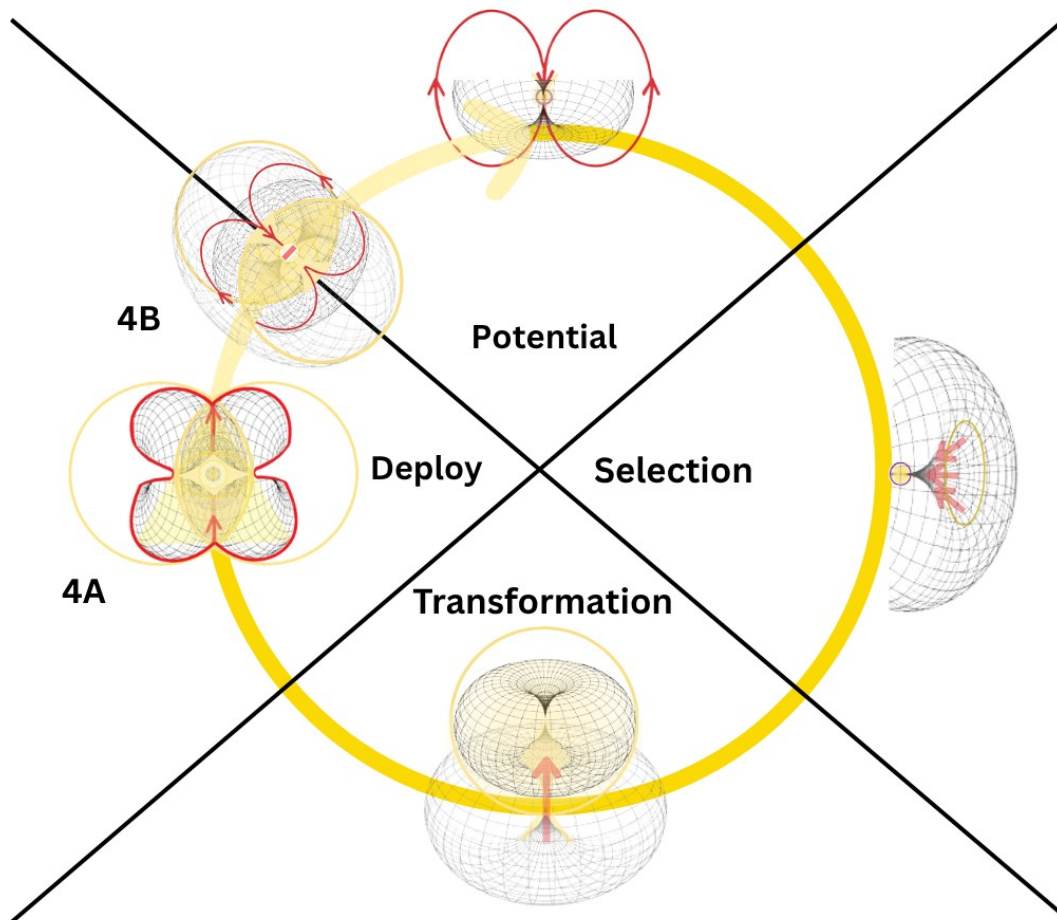


Figure 1. The First Pattern Information Emergence Cycle (IEC). The lifecycle proceeds through four quadrants: Potential (Q1), Selection (Q2), Transformation (Q3), and Deploy (Q4), with Q4 bifurcating into continuation (Q4A) and reorientation (Q4B). Each quadrant comprises two information operators (entry and exit) that together produce the quadrant's function.

Post-failure reorientation targets systematically novel options

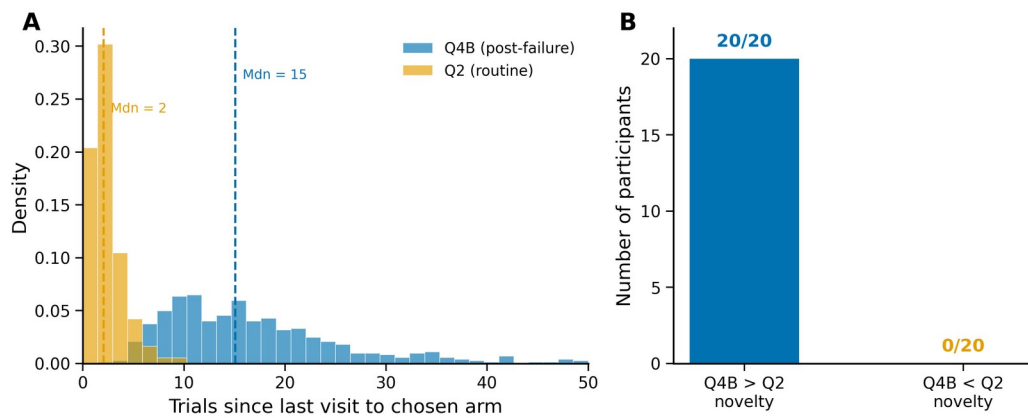


Figure 2. Post-failure reorientation targets systematically novel options. Q4B-triggered transitions select arms unvisited for a median of 15 trials versus 2 trials for routine exploration ($p = 3.9 \times 10^{-83}$).

Cross-domain unidirectional quadrant flow

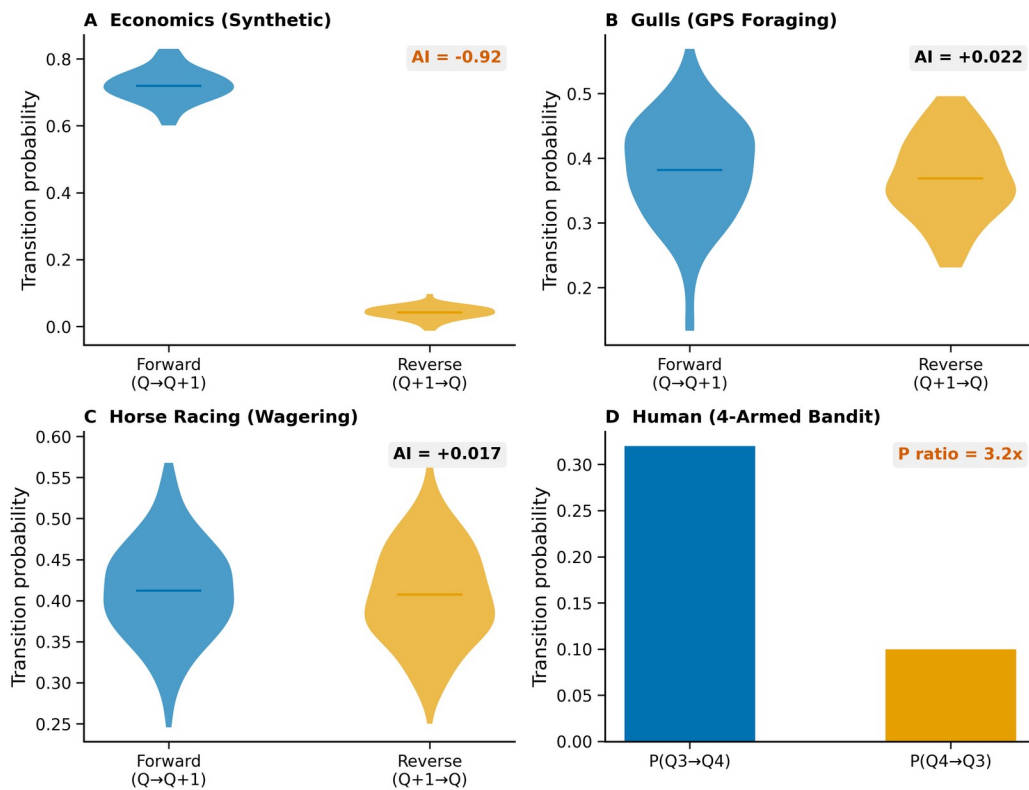


Figure 3. Cross-domain unidirectional flow asymmetry across macroeconomic, avian, mammalian, and human domains.

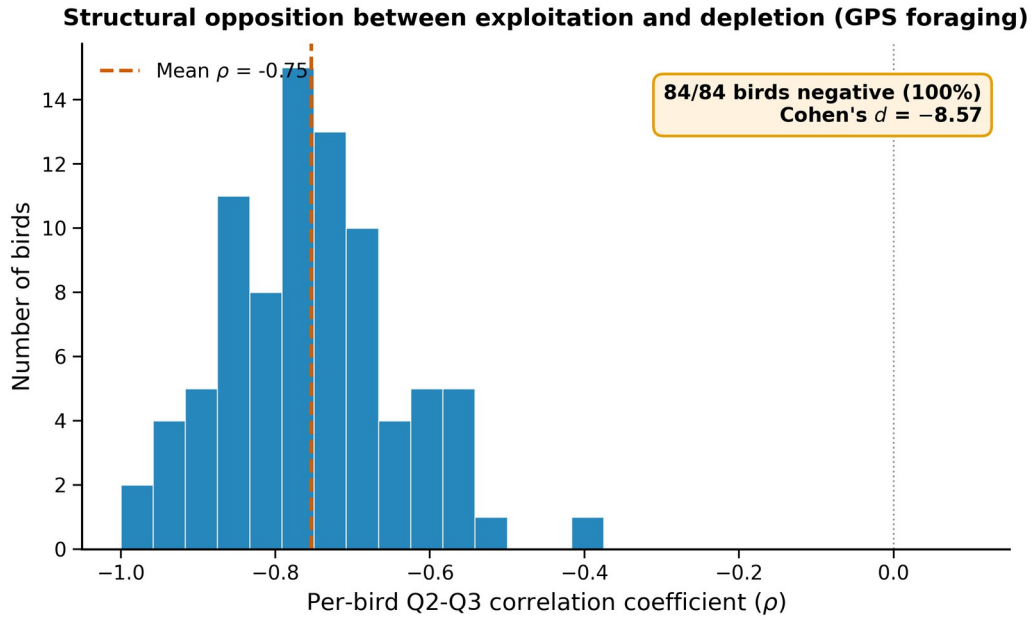


Figure 4. Structural opposition between selection (Q2) and transformation (Q3) in avian foraging. All 84 individual birds show negative Q2-Q3 correlation.

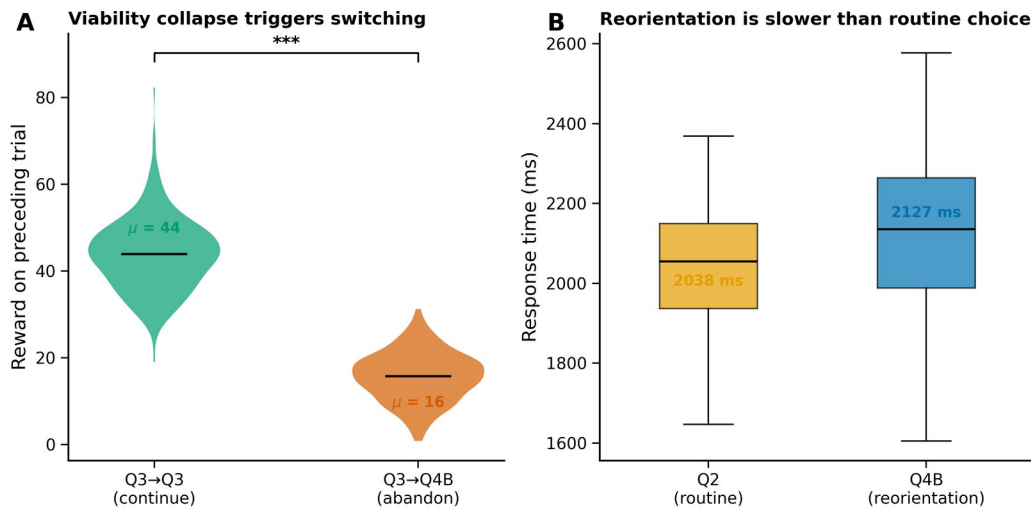


Figure 5. Viability gating mechanism. Reward distribution before exploitation continuation versus abandonment, and response time comparison.